2.1 Introducing Astra

Our conversational app is an AI persona called Astra. An AI's *persona* defines the unique style, characteristics, and behavioral traits the application exhibits—like Google's Gemini. You can see a screenshot of Astra at work in figure 2.1. In this interaction, a user types a simple query, "Tell me a dad joke." In a matter of seconds, our application, which is powered by an LLM and a few lines of code, processes the request and generates a humorous response.

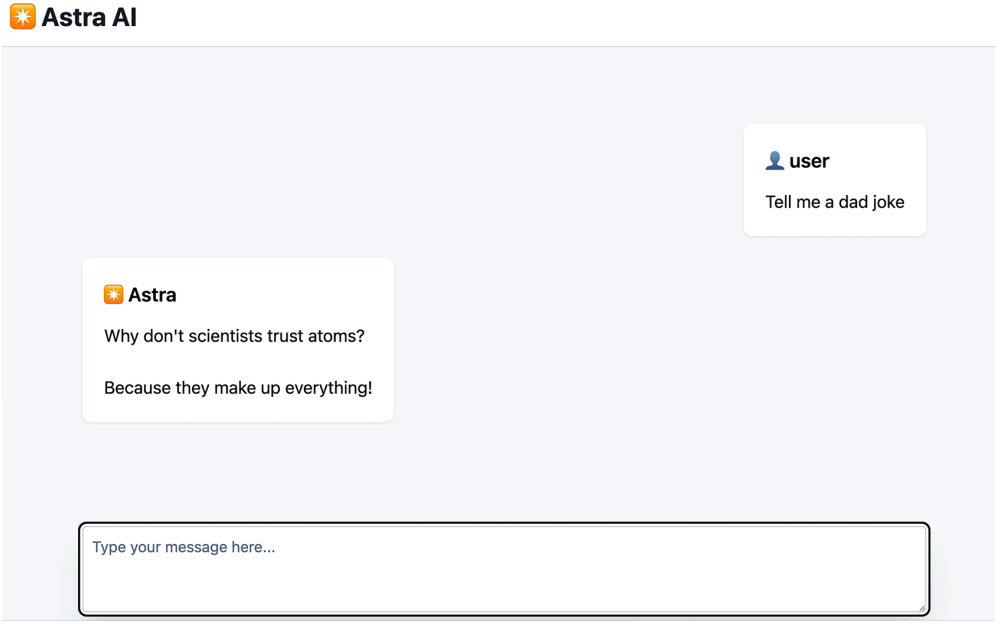


Figure 2.1 A simple conversational UI demonstrating a user query and an LLM-generated response

For our Astra app, we'll use pretrained models via public APIs such as Gemini and OpenAI, which offer a practical way to access state-of-the-art language models without the complexity of hosting and maintaining them ourselves. In fact, we will review two implementations: one built with React and Express.js, and one built with Next.js, with the goal of improving the project in subsequent chapters. By comparing these approaches, you'll gain valuable insights into the strengths and tradeoffs of each framework.

> **NOTE** For step-by-step instructions on how to create and configure API keys for both OpenAI and Google Gemini, including important billing considerations and free tier availability, refer to the appendix.